

# A Comparative Study of 50-Millesimal Potencies and Centesimal Potencies in the Treatment of Tinea Corporis: An Open, Randomised, Pilot Trial

Pallavi Hazra<sup>1</sup> Shyamal Kumar Mukherjee<sup>2</sup> Subhasish Ganguly<sup>1</sup> Aniket Singha Roy<sup>1</sup> Souvik Dutta<sup>1</sup>   
Satarupa Sadhukhan<sup>3</sup>  Munmun Koley<sup>4</sup> Subhranil Saha<sup>5</sup>

<sup>1</sup> Department of Organon of Medicine and Homoeopathic Philosophy, D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, India

<sup>2</sup> D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, India

<sup>3</sup> Department of Organon of Medicine and Homoeopathic Philosophy, National Institute of Homoeopathy, Kolkata, West Bengal, India

<sup>4</sup> Department of Organon of Medicine and Homoeopathic Philosophy, State National Homoeopathic Medical College and Hospital, Lucknow, Uttar Pradesh, India

Address for correspondence Pallavi Hazra, MD (Hom), Department of Organon of Medicine and Homoeopathic Philosophy, D. N. De Homoeopathic Medical College and Hospital, 12, Gobinda Khatick Road, Kolkata 700046, West Bengal, India (e-mail: pallavihazra1991@gmail.com).

<sup>5</sup> Department of Repertory, D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, India

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## Abstract

**Introduction** Standard management of dermatophytosis involves the use of topical anti-fungals in limited disease and oral therapy for more extensive cases. Despite the evidences favouring homeopathy in tinea corporis (TC) is inadequate, patients are increasingly opting for it. This trial attempts to assess the feasibility of a trial evaluating individualised homeopathic (IH) medicines in centesimal and 50-millesimal potencies in treating TC.

**Methods** An open, randomised, pilot feasibility trial was conducted at the dermatology outpatient of D. N. De Homoeopathic Medical College and Hospital, West Bengal, India. Patients were randomised to receive IH medicines in either centesimal potencies ( $n = 30$ ) or in 50-millesimal potencies ( $n = 30$ ). Primary outcome measure was intensity of pruritus on a 0 to 10 numeric rating scale; secondary outcomes were Skindex-29 and dermatology life quality index questionnaires—all measured at baseline, and after 6 and 12 weeks of intervention. Group differences were calculated on intention-to-treat (ITT) sample as well as following per protocol (PP) analysis.

**Results** Recruitment and attrition rates were 69 and 16.7% respectively. Both ITT and PP analysis revealed that intra-group changes in both the primary and secondary outcomes to be statistically significant in both the groups (all  $p < 0.05$ , one-way repeated measure analysis of variance). There were no significant group differences over 6 and 12 weeks (all  $p > 0.05$ , unpaired  $t$ -tests); thus, indicating similar pattern of improvement in both the groups. Most frequently used medicines were *Natrum sulphuricum* (28.3%), *Natrum muriaticum* (25%), *Sulphur* (8.3%) and *Bacillinum* (6.7%). No harms, unintended effects, homeopathic aggravations or any serious adverse events were reported from either group.

**Conclusion** An adequately powered trial exploring effectiveness of both scales in TC appeared to be feasible in future. Though the sample size was too small to arrive at any definite conclusion, both centesimal and 50-millesimal potencies of IH medicines appeared to be equally useful in improving pruritus intensity and quality of life in patients suffering from TC.

## Keywords

- centesimal potency
- 50-millesimal potency
- homeopathy
- pilot trial
- randomised controlled trials
- tinea corporis

## Introduction

Superficial mycotic infection is the most common infection worldwide. About 20 to 25% of the world population are having some form of fungal infection, which is usually anthropophilic trichophyton infection.<sup>1</sup> In the tropical countries, dermatophytoses are more common.<sup>2</sup> In India, a recent study has detected *Trichophyton verrucosum* to be the most common dermatophyte that was followed by *T. rubrum* and *T. mentagrophytes*.<sup>3</sup> *T. verrucosum* was the most common organism in primary cases of tinea cruris and tinea corporis (TC; ringworm).<sup>3</sup> In West Bengal, TC has been found to be the most common superficial fungal infection.<sup>4</sup>

A typical TC lesion is a red, annular, scaly, pruritic patch with central clearance, and having an active border. The lesions may be solitary as well as multiple. Usually, the size ranges from 1 to 5 cm. Several other skin disorders (eczema, psoriasis, and seborrhoeic dermatitis) may be misdiagnosed as TC. Appearance of the lesion is often sufficient to diagnose TC, tinea cruris, and tinea pedis. In cases of atypical presentation, potassium hydroxide (KOH) preparation or culture may be performed for diagnosis.<sup>5</sup> Topical antifungal agents are commonly used for treatment. Oral therapies may be indicated for extensive disease, failed topical treatment, immune-compromised patients or severe moccasin-type tinea pedis.<sup>6</sup> In a case-control study,<sup>7</sup> a positive association was found between chronic and chronic-relapsing tinea and pre-existing diabetes mellitus, family history of tinea and personal history of cooking food. A systemic review<sup>8</sup> assessed the evidence for the effectiveness and safety of topical treatments for tinea cruris and TC. Most of the 129 randomised controlled trials (RCTs) assessed in this review were poorly designed and inadequately reported. Therefore, RCTs with sound methodology are much needed to give a clear idea of treatment regarding dose and duration of therapy.<sup>9</sup> In a systematic review<sup>10</sup> of controlled clinical trials with homeopathic interventions, trials investigating treatments for cutaneous diseases were found to be very limited. Most of the trials with interpretable results did not indicate any positive effects in favour of homeopathy. Only three trials showed positive effects, but suffered from low methodological qualities. Another clinical study<sup>11</sup> conducted on 36 patients aimed to verify whether tinea was curable by its pathological similimum in high potencies. Size, shape and itching were assessed with comparison to the original lesions. Improvement in itching was noted in 22 cases.

Dr Samuel Hahnemann introduced the process of potentisation in homeopathy. In the 50-millesimal scale, the 1st potency contains 1/50,000th part of the original drug substance. The next potency contains 1/50,000th part of the 1st potency and so on. Likewise, in the centesimal scale, the 1st potency contains 1/100th part of the original drug substance. The 2nd potency contains 1/100th part of the 1st potency and so on.<sup>12</sup>

In this external pilot study, we aimed to explore the feasibility of a definitive and adequately powered trial of individualised homeopathic (IH) medicines in different scales of potencies in the treatment of TC. We tried to address feasibility issues like screening, recruitment and retention

rates and suitability of the outcome measures and to inform sample size for the main trial. We also tried to estimate preliminary indications, difference between centesimal and 50-millesimal potencies in treatment of TC, if any, and to find out the commonly indicated medicines for such clinical entity.

## Methods

**Trial design:** This prospective, open-label, randomised, two parallel arms, pilot trial was conducted at the outpatient department of D. N. De Homoeopathic Medical College and Hospital, 12, Gobinda Khatick Road, Kolkata 700046, West Bengal. The study protocol was approved by the Institutional Ethical Committee (IEC) vide reference no. DHC/Aca(PG)-29/14/861/18, dated 13<sup>th</sup> September 2018. The trial protocol and full dissertation were submitted as the postgraduate thesis of the corresponding author to The West Bengal University of Health Sciences, DD36, 2<sup>nd</sup> Avenue, DD Block, Sector I, Bidhan Nagar, Kolkata 700064, West Bengal. We had a prospective registration of the trial protocol in clinical trial registry—India (Trial registration: CTRI/2018/10/016065) with a secondary identifier, that is, Universal Trial Number (UTN: U1111-1221-8309).

**Participants:** Included were the patients having infection in the body excluding scalp, feet, groin, nail, with the symptoms of ring shaped lesions with advancing scaly border and central clearing or scaly patches with a distinct border with positive diagnosis of tinea made by microscopic examination of scrapings of affected skin after forming KOH solution and culture (2020 International Classification of Diseases, Tenth Revision, Clinical Modification diagnosis code B35.4).<sup>13</sup> Patients aged between 18 and 60 years, of either sex, literate patients who could read Bengali, gave written consent to participate in the study and willing to take only homeopathic medicines were included in the study. Excluded were the patients suffering from uncontrolled systemic illness or life-threatening infections, advanced pathological conditions, emergency conditions, cases already undergoing treatment of any system either for TC or for any other chronic disease, substance abuse and/or dependence, pregnant, puerperal or lactating women, patients with psychiatric diseases and self-reported immune-compromised states.

**Intervention:** Patients were divided into two parallel arms—

1. Centesimal potency group: Individualised homeopathic (IH) medicines were prescribed in centesimal potency. Treatment was started from 30C or 200C, as appropriate to the case or condition, and gradually increased as per homeopathic principles. Each dose consisted of a single drop of the indicated medicine (preserved in 90% v/v ethanol) in 5 mL of distilled water. Dosage regime was once daily for 1 to 2 days, followed by placebo once daily; up to reassessment after 1 month and then adjusted, if necessary.
2. 50-millesimal potency group: IH medicine was given in 50-millesimal potency to the patients of this group. In 50-millesimal scale, a single-medicated cane sugar globules

of poppy seed size (no. 10) were dissolved in 80 mL distilled water with addition of two drops of 90% v/v ethanol; 16 doses marked on the vial; each dose of 5 mL was to be taken after 10 uniformly forceful downward strokes to the vial in 45 mL normal water in a clean cup, to stir well, to take 5 mL of this liquid orally and to discard rest of the liquid from the cup. The potencies started from 0/1 once daily, and the numbers of required doses were tailored as per individual requirement of the cases. Potencies were gradually increased depending on the outcome and as per homeopathic principles.

All the medicines were procured from a Good Manufacturing Practice-certified firm—Hahnemann Publishing Company Private Ltd., 165, Bipin Bihari Ganguly Street, Baithakkhana, Bowbazar, Kolkata, West Bengal 700012. Final selection of the single individualised medicine and dosage was in accordance with the standard homeopathic guidelines and agreement among three homeopaths. The prescriptions on follow-up visits were generated as per relevant homeopathic principles and were recorded in follow-up sheets. Homeopathic remedies were permitted to change during the study as and when required in adherence to the homeopathic principles. Two of the prescribers possessed master's degree in homeopathy with more than 20 years of experience of teaching and practicing classical homeopathy and the rest were postgraduate trainees at the said hospital. All the homeopaths involved were affiliated with state councils.

**General management:** The patients were asked to avoid wearing tight clothes, sharing clothes, bedding, towels with other, exposure to heat and humidity.

## Outcomes

1. Primary outcome—Numeric Rating Scale (NRS) 0–10, where number '0' represented 'no itch' and number '10' represented 'worst imaginable itch'. Subjects were asked to rate the intensity of their itch using this scale. It had high reliability and concurrent validity and was a popular choice for all patients due to its simple format.<sup>14</sup>
2. Secondary outcome(s)—
  - (a) Skindex-29 questionnaire—The Skindex is a health-related quality of life instrument designed to measure the effects of skin disease on patients' lives.<sup>15</sup> This dermatology-specific questionnaire has been extensively studied and refined in different population samples. The questionnaire covers areas considered crucial in an instrument designed to evaluate quality of life, such as: degree of symptoms, psychosocial functioning, and emotional status (Suppl. file 1).
  - (b) Dermatology life quality index (DLQI) questionnaire—It is a valid questionnaire including 10 items on the patient's symptoms, feelings, routine activities, kind of clothes, social or leisure, activities, physical exercise, educational activities, sexual activities, interpersonal relationships and treatment options with 0 to 30 points score.<sup>16</sup> (Suppl. file 2). The questions were classified into six headings: symptoms and feelings

(questions 1 and 2), daily activities (questions 3 and 4), leisure (questions 5 and 6) and personal relationships (questions 8 and 9) each item with maximum score 6; work and school (question 7) and treatment (question 10) each item with maximum score of 3.

All of these three outcomes were assessed at baseline, after 6 weeks, and after 12 weeks of intervention. All the primary and secondary parameters were patients' self-rated. We also had provision to record the (serious) adverse events and intercurrent illnesses.

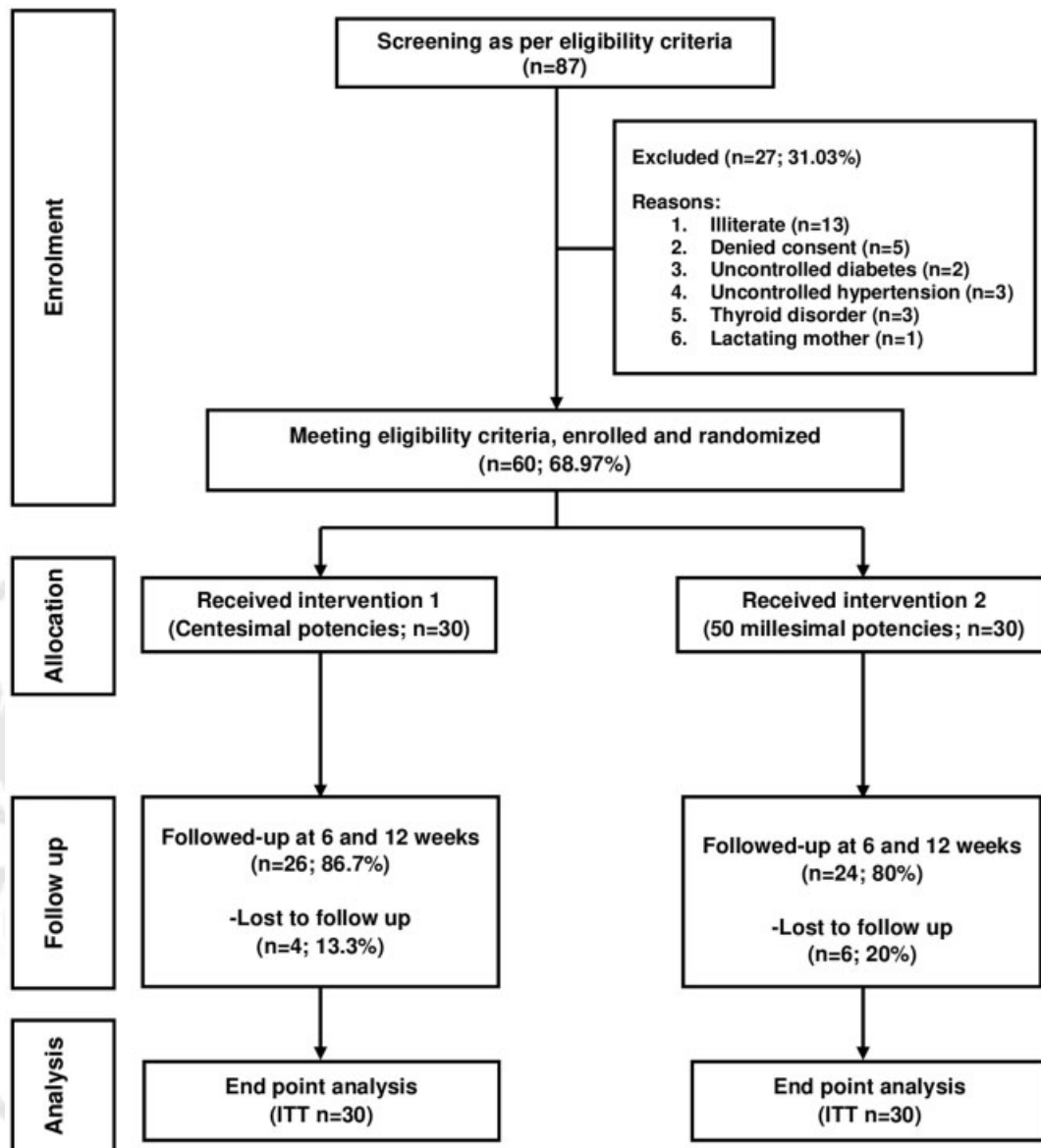
**Sample size:** Formal effect size and subsequent sample size calculation were not possible on account of absence of any earlier study of this explanatory design. Keeping in mind the time constraint and feasibility issues, a target sample size of 60 (30 in each group) deemed to be attainable in this trial.

**Randomisation:** A random number chart containing 60 random numbers in 1:1 ratio was generated by using a StatTrek random number generator using restricted 6 blocks of size 10 ( $6 \times 10 = 60$ ); thus, equal numbers of patients were randomised to either 50-millesimal ( $n = 30$ ) or centesimal group ( $n = 30$ ).

**Blinding:** Due to dissimilarity in dispensing of medicines in centesimal and 50-millesimal potencies, blinding was not possible. Hence, blinding method was not adopted. The randomisation chart was made available to the physicians and pharmacists to prescribe and dispense medicines accordingly.

**Statistical methods:** The study results were analysed following intention-to-treat (ITT) as well as per protocol (PP) approach. Baseline comparability in sociodemographic features and outcome measures between groups was checked either by chi-squared tests (for categorical variables) or by unpaired *t*-tests (for continuous variables). Baseline differences were adjusted by analysis of covariance (ANCOVA) models. Group differences (independent observations) were tested after 6 and 12 weeks by unpaired *t*-tests. Dependent observations of continuous outcomes at baseline and at different points of time were compared using one-way repeated measure analysis of variance (ANOVA). A *p*-values were set at less than 0.05 two-tailed as statistically significant. Statistical analysis was performed using Statistical Package for the Social Sciences, version 23.0 (IBM Corp., IBM SPSS Statistics for Windows, Armonk, New York, United States).

**Ethical considerations:** Irrespective of the assigned codes, adverse events were recorded and treated accordingly as per homeopathic principles. In this project, no new drug was experimented nor was any new treatment protocol adopted. Intervention was in strict adherence to classical homeopathic treatment. Prior to enrolment, each patient was provided with a patient information sheet in local vernacular Bengali detailing the objectives, methods, risks and benefits of participating, and confidentiality issues. Subsequent to that, written informed consent was obtained. Approval was taken from the IEC prior to initiation. The protocol conformed to the declaration of Helsinki for ethical conduct of clinical trials involving human participants.



**Fig. 1** Consolidated Statement for Reporting Trials (CONSORT) study flow diagram.

**Reporting of adverse events:** The investigators instructed the patients to report any harm, unintended effect, serious adverse event and undue aggravation; either directly in the outpatient departments or over phone during the trial. But no such cases were reported.

**Trial reporting:** Reporting of the trial was in compliance with the Consolidated Statement for Reporting Trials (CONSORT)<sup>17</sup> and Reporting Data on Homeopathic Treatment (RedHot)<sup>18</sup> guidelines (Suppl. files 3 and 4: CONSORT and RedHot checklists).

## Results

**Study flow:** As per the pre-specified inclusion and exclusion criteria, 87 patients suffering from TC were screened; out of which, 27 were excluded on account of varied reasons (listed

in ► **Fig. 1**); remaining 60 met the eligibility criteria and were enrolled. Baseline socio-demographic and outcome data were obtained for these 60 enrolled patients and were subjected to IH medicine, either in 50-millesimal potency or centesimal potency as per the random number chart. The outcome data was recorded at baseline before intervention and at the intervals of 6 and 12 weeks. During the course of treatment, 10 patients dropped out (6 from 50-millesimal and 4 from centesimal group); 50 patients have completed the trial (► **Fig. 1**).

**Study duration and follow-up:** The study continued for 14 months (December 2018–January 2020), whereas the duration of therapy for each patient was 3 months.

**Numbers analysed:** All the randomised patients ( $n = 60$ ) entered into the final analysis as per ITT analysis, that is, 30 in each group. PP analysis was also performed with 50 patients.



**Table 1** Comparison of socio-demographic characteristics between two groups at baseline ( $n = 60$ )

| Features                                | Centesimal group<br>( $n = 30$ ) | 50-millesimal group<br>( $n = 30$ ) | p-Value |
|-----------------------------------------|----------------------------------|-------------------------------------|---------|
| Age (y) <sup>a</sup>                    | 35.9 ± 12.9                      | 33.0 ± 10.0                         | 0.310   |
| Sex <sup>b</sup>                        |                                  |                                     |         |
| ■ Male                                  | 7 (23.3)                         | 11 (36.7)                           | 0.398   |
| ■ Female                                | 23 (76.7)                        | 19 (63.3)                           |         |
| Duration of suffering (mo) <sup>a</sup> | 14.6 ± 2.7                       | 12.9 ± 2.4                          | 0.955   |
| Number of lesions                       | 1.5 ± 0.7                        | 2.1 ± 1.0                           | 0.011*  |
| Size of lesions (diameter; cm)          | 3.7 ± 1.3                        | 3.7 ± 1.6                           | 0.858   |
| Site of lesions(s) <sup>b</sup>         |                                  |                                     |         |
| ■ Abdomen                               | 14 (46.7)                        | 8 (26.7)                            | 0.156   |
| ■ Axilla                                | 6 (20)                           | 9 (30)                              |         |
| ■ Breast                                | 3 (10)                           | 5 (16.7)                            |         |
| ■ Face                                  | 1 (3.3)                          | 2 (6.7)                             |         |
| ■ Hand                                  | 4 (13.3)                         | 5 (16.7)                            |         |
| ■ Others                                | 2 (6.7)                          | 1 (3.3)                             |         |
| Treatment taken <sup>b</sup>            |                                  |                                     |         |
| ■ None                                  | 2 (6.7)                          | 3 (10)                              | 0.902   |
| ■ Allopathy                             | 25 (83.3)                        | 27 (90)                             |         |
| Body mass index <sup>a</sup>            | 23.7 ± 3.3                       | 24.8 ± 3.3                          | 0.599   |
| Blood pressure (mm Hg) <sup>a</sup>     |                                  |                                     |         |
| ■ Systolic                              | 117.3 ± 10.1                     | 118.2 ± 8.7                         | 0.724   |
| ■ Diastolic                             | 77.3 ± 7.4                       | 77.3 ± 7.8                          | 1.000   |
| Residence <sup>b</sup>                  |                                  |                                     |         |
| ■ Rural                                 | 9 (30)                           | 8 (26.7)                            | 1.000   |
| ■ Urban                                 | 21 (70)                          | 22 (73.3)                           |         |
| Marital status <sup>b</sup>             |                                  |                                     |         |
| ■ Single                                | 7 (23.3)                         | 7 (23.3)                            | 1.000   |
| ■ Married                               | 23 (76.7)                        | 23 (76.7)                           |         |
| Educational status <sup>b</sup>         |                                  |                                     |         |
| ■ 8 <sup>th</sup> std. or less          | 19 (63.3)                        | 20 (66.7)                           | 0.791   |
| ■ 9–12 <sup>th</sup> std.               | 4 (13.3)                         | 5 (16.7)                            |         |
| ■ Graduate or above                     | 7 (23.3)                         | 5 (16.7)                            |         |
| Employment status <sup>b</sup>          |                                  |                                     |         |
| ■ Business                              | 2 (6.7)                          | 1 (3.3)                             | 0.832   |
| ■ Service                               | 14 (46.7)                        | 15 (50)                             |         |
| ■ Dependent                             | 14 (47.7)                        | 14 (46.7)                           |         |
| Family income status <sup>b</sup>       |                                  |                                     |         |
| ■ Poor                                  | 24 (80)                          | 24 (80)                             | 1.000   |
| ■ Middle                                | 4 (13.3)                         | 4 (13.3)                            |         |
| ■ Affluent                              | 2 (6.7)                          | 2 (6.7)                             |         |

<sup>a</sup>Continuous data presented as means ± standard deviations and unpaired *t* test applied; *t*<sub>58</sub>: *t* score at 58 degrees of freedom; <sup>b</sup>Categorical data presented as absolute values (percentages) and chi-squared test applied; \**p*-values less than 0.05 two-tailed were considered as statistically significant.

Baseline descriptive statistics: Fourteen variables were studied across the two treatment groups—age, sex, duration of suffering, number of lesions, size of lesions, site of lesions, treatment taken, body mass index, blood pressure, residence, marital status, education, employment status, family income status to check whether the distribution of the variables between the two groups were statistically different or not,

**Table 2** Comparison of intensity of pruritus (0–10 NRS) at baseline and over 6 and 12 weeks

| Intensity of pruritus<br>(0–10 NRS) | Baseline:<br>Mean (SD) | After 6 weeks:<br>Mean (SD) | After 12 weeks:<br>Mean (SD) | Wilks'<br>lambda | Partial<br>eta squared | <i>F</i> <sub>2, 28</sub> | <i>p</i> -Value <sup>b</sup> |
|-------------------------------------|------------------------|-----------------------------|------------------------------|------------------|------------------------|---------------------------|------------------------------|
| ■ Centesimal group                  | 8.1 ± 1.9              | 5.8 ± 3.0                   | 3.9 ± 3.8                    | 0.415<br>0.496   | 0.585<br>0.504         | 19.745<br>14.211          | <0.001*<br><0.001*           |
| ■ 50-millesimal group               | 8.0 ± 1.6              | 5.7 ± 3.0                   | 4.4 ± 3.9                    |                  |                        |                           |                              |
| Mean group difference<br>(SE)       | 0.0 (0.5)              | 0.1 (0.8)                   | −0.4 (1.0)                   |                  |                        |                           |                              |
| 95% CI                              | −0.9, 0.9              | −1.4, 1.6                   | −2.4, 1.6                    |                  |                        |                           |                              |
| <i>t</i> <sub>58</sub>              | 0.073                  | 0.130                       | −0.437                       |                  |                        |                           |                              |
| <i>p</i> -Value <sup>a</sup>        | 0.942                  | 0.897                       | 0.664                        |                  |                        |                           |                              |

Abbreviations: ANOVA, analysis of variance; ANCOVA, analysis of covariance; CI, confidence interval; NRS, numeric rating scale; SD, standard deviation; SE, standard error.

<sup>a</sup>Unpaired *t* tests; *t*<sub>58</sub>: *t* score at 58 degrees of freedom; <sup>b</sup>One way repeated measure analysis of variance. \**p* < 0.05 considered as statistically significant; *p*-value reflects inter-group differences detected by unpaired *t*-tests, whereas *p*-value<sup>b</sup> represents intra-group changes detected by one-way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to *R*<sup>2</sup> from multiple linear regressions.

( $n = 60$ ; centesimal group: 30, 50-millesimal group: 30).

(Adjusted for baseline differences in number of lesions by ANCOVA model; intention to treat analysis).

**Table 3** Comparison of intensity of pruritus (0–10 NRS) at baseline and over 6 and 12 weeks

| Intensity of pruritus (0–10 NRS) | Baseline: Mean (SD) | After 6 weeks: Mean (SD) | After 12 weeks: Mean (SD) | Wilks' lambda  | Partial eta squared | $F_{2, 24}$ or $F_{2, 22}$ | $p$ -Value <sup>b</sup> |
|----------------------------------|---------------------|--------------------------|---------------------------|----------------|---------------------|----------------------------|-------------------------|
| ■ Centesimal group               | 8.1 ± 1.9           | 5.5 ± 3.0                | 3.3 ± 3.6                 | 0.430<br>0.413 | 0.570<br>0.587      | 15.927<br>15.633           | <0.001*<br><0.001       |
| ■ 50-millesimal group            | 8.0 ± 1.6           | 5.2 ± 2.9                | 3.4 ± 3.7                 |                |                     |                            |                         |
| Mean group difference (SE)       | 0.0 (0.5)           | 0.3 (0.8)                | -0.1 (1.0)                |                |                     |                            |                         |
| 95% CI                           | -0.9, 0.9           | -1.3, 2.0                | -2.2, 1.9                 |                |                     |                            |                         |
| $t_{58}$                         | 0.073               | 0.399                    | -0.145                    |                |                     |                            |                         |
| $p$ -Value <sup>a</sup>          | 0.942               | 0.692                    | 0.885                     |                |                     |                            |                         |

Abbreviations: ANOVA, analysis of variance; ANCOVA, analysis of covariance; CI, confidence interval; NRS, numeric rating scale; SD, standard deviation; SE, standard error.

<sup>a</sup>Unpaired  $t$  tests;  $t_{58}$ :  $t$  score at 58 degrees of freedom; <sup>b</sup>One-way repeated measure analysis of variance, \* $p < 0.05$  considered as statistically significant;  $p$ -value <sup>a</sup> reflects inter-group differences detected by unpaired  $t$ -tests, whereas  $p$ -value <sup>b</sup> represents intra-group changes detected by one-way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to  $R^2$  from multiple linear regressions.

$n = 60$ ; centesimal group: 30, 50-millesimal group: 30.

Adjusted for baseline differences in number of lesions by ANCOVA model; per protocol analysis.

**Table 4** Comparison of the Skindex-29 scores at baseline and over 6 and 12 weeks

| Skindex-29 scores                                                                                                                                 | Baseline: Mean (SD)                                                       | After 6 weeks: Mean (SD)                                                | After 12 weeks: Mean (SD)                                                  | Wilks' lambda  | Partial eta squared | $F_{2, 28}$      | $p$ -Value <sup>b</sup> |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------|---------------------|------------------|-------------------------|
| Emotion (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup>     | 67.6 ± 15.9<br>73.7 ± 16.9<br>-6.1 (4.2)<br>-14.6, 2.4<br>-1.430<br>0.158 | 55.3 ± 25.0<br>51.3 ± 22.8<br>4.1 (6.2)<br>-8.3, 16.4<br>0.657<br>0.514 | 44.9 ± 29.1<br>47.4 ± 28.8<br>-2.5 (7.5)<br>-17.5, 12.4<br>-0.339<br>0.736 | 0.566<br>0.531 | 0.434<br>0.469      | 10.743<br>12.375 | <0.001***<br><0.001***  |
| Symptoms (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup>    | 62.3 ± 14.3<br>65.8 ± 12.4<br>-3.5 (3.5)<br>-10.5, 3.4<br>-1.019<br>0.312 | 50.8 ± 18.1<br>49.8 ± 18.3<br>1.0 (4.7)<br>-8.4, 10.4<br>0.206<br>0.838 | 42.6 ± 23.3<br>43.5 ± 22.2<br>-1.0 (5.9)<br>-12.7, 10.8<br>-0.162<br>0.872 | 0.567<br>0.459 | 0.433<br>0.541      | 10.710<br>16.521 | <0.001***<br><0.001***  |
| Functioning (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup> | 60.6 ± 17.6<br>63.2 ± 17.1<br>-2.6 (4.5)<br>-11.5, 6.4<br>-0.569<br>0.572 | 49.3 ± 23.3<br>47.4 ± 22.1<br>1.9 (5.9)<br>-9.8, 13.6<br>0.748<br>0.322 | 42.2 ± 26.9<br>45.2 ± 26.2<br>-3.0, 6.9<br>-16.7, 10.7<br>-0.438<br>0.663  | 0.641<br>0.646 | 0.359<br>0.354      | 7.853<br>7.684   | 0.002**<br>0.002**      |
| Overall (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup>     | 63.4 ± 14.7<br>67.4 ± 14.0<br>-4.0 (3.7)<br>-11.4, 3.4<br>-1.081<br>0.284 | 51.7 ± 22.0<br>49.3 ± 20.8<br>2.4 (5.5)<br>-8.7, 13.5<br>0.436<br>0.665 | 43.2 ± 26.4<br>45.5 ± 25.8<br>-2.3 (6.7)<br>-15.8, 11.1<br>-0.349<br>0.729 | 0.579<br>0.537 | 0.421<br>0.463      | 10.172<br>12.075 | <0.001***<br><0.001***  |

Abbreviations: ANOVA, analysis of variance; ANCOVA, analysis of covariance; CI, confidence interval; SD, standard deviation; SE, standard error.

<sup>a</sup>Unpaired  $t$  tests;  $t_{58}$ :  $t$  score at 58 degrees of freedom; <sup>b</sup>One-way repeated measure analysis of variance;  $p < 0.05$  considered as statistically significant; \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ ;  $p$ -value <sup>a</sup> reflects inter-group differences detected by unpaired  $t$ -tests, whereas  $p$ -value <sup>b</sup> represents intra-group changes detected by one-way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to  $R^2$  from multiple linear regressions.

$n = 60$ ; centesimal group: 30, 50-millesimal group: 30.

Adjusted for baseline differences in number of lesions by ANCOVA model; intention to treat analysis.

by using unpaired *t*-test and chi-squared test for continuous and categorical variables respectively. No significant differences were detected between groups (all  $p > 0.05$ ), except the fact that the number of lesions was significantly higher in the 50-millesimal group than centesimal ( $p = 0.011$ ) (►Table 1). These baseline differences of number of lesions were adjusted by ANCOVA model.

Statistical analysis and interpretation:

1. Comparison of intensity of itching NRS score: As per ITT analysis, over 6 and 12 weeks of intervention in either of the two groups, there were significant intra-group improvement in intensity of pruritus in both groups ( $p < 0.001$ , one-way repeated measure ANOVA) without any significant group differences (week 6:  $p = 0.897$ , week 12:  $p = 0.664$ ). Thus, similar pattern of improvement was observed in both the groups in intensity of pruritus (►Table 2). In PP analysis, similar pattern was observed, where there were significant intra-group improvement in intensity of pruritus in both groups ( $p < 0.001$ , one-way repeated measure ANOVA) without any significant group differences (week 6:  $p = 0.692$ , week 12:  $p = 0.885$ ) (►Table 3).

Intensity of pruritus was found to be increased in five patients in the 50-millesimal group after 6 weeks of therapy and in two patients in the centesimal group after same duration.

2. Comparison of the Skindex-29 scores: As per ITT analysis, there were no significant group differences (all  $p > 0.05$ , unpaired *t*-tests) in distribution of the Skindex-29 overall score and three subscale scores at baseline; thus, ensuring comparability between the two groups. Significant intra-group improvement (all  $p < 0.05$ , one-way repeated measure ANOVA) was detected in all the subscales and overall scores over time (►Table 4). In PP analysis, similar pattern was observed over time, where significant intra-group improvement (all  $p < 0.05$ , one-way repeated measure ANOVA) was detected in all the subscales and overall scores (►Table 5).

3. Comparison of the DLQI scores: No significant baseline group differences existed (all  $p > 0.05$ , unpaired *t*-tests), thus ensuring comparability between the two groups. Again, as per ITT analysis, significant intra-group improvement (all  $p < 0.05$ , one-way repeated measure ANOVA) was found over time in all the subscale scores (except 50-millesimal scale of treatment domain) and

**Table 5** Comparison of the Skindex-29 scores at baseline and over 6 and 12 weeks

| Skindex-29 scores                                                                                                                                 | Baseline:<br>Mean (SD)                                                    | After 6 weeks:<br>Mean (SD)                                             | After 12 weeks:<br>Mean (SD)                                               | Wilks' lambda  | Partial eta squared | $F_{2, 28}$      | $p$ -Value <sup>b</sup> |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------|---------------------|------------------|-------------------------|
| Emotion (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup>     | 67.6 ± 15.9<br>73.7 ± 16.9<br>–6.1 (4.2)<br>–14.6, 2.4<br>–1.430<br>0.158 | 53.4 ± 25.0<br>46.4 ± 21.8<br>7.0 (6.6)<br>–6.4, 20.3<br>1.047<br>0.300 | 41.3 ± 28.5<br>41.6 ± 28.4<br>–0.3 (8.1)<br>–16.5, 15.9<br>–0.034<br>0.973 | 0.579<br>0.413 | 0.421<br>0.587      | 8.738<br>15.640  | 0.001**<br><0.001***    |
| Symptoms (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup>    | 62.3 ± 14.3<br>65.8 ± 12.4<br>–3.5 (3.5)<br>–10.5, 3.4<br>–1.019<br>0.312 | 48.1 ± 17.5<br>46.2 ± 18.7<br>1.9 (5.1)<br>–8.3, 12.2<br>0.382<br>0.704 | 38.7 ± 22.2<br>38.3 ± 21.9<br>0.3 (6.2)<br>–12.2, 12.9<br>0.055<br>0.956   | 0.524<br>0.417 | 0.476<br>0.583      | 10.904<br>15.379 | <0.001***<br><0.001***  |
| Functioning (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup> | 60.6 ± 17.6<br>63.2 ± 17.1<br>–2.6 (4.5)<br>–11.5, 6.4<br>–0.569<br>0.572 | 47.9 ± 22.9<br>42.8 ± 21.4<br>5.1 (6.3)<br>–7.5, 17.7<br>0.812<br>0.421 | 39.7 ± 26.4<br>40.1 ± 26.1<br>–0.3 (7.4)<br>–15.3, 14.6<br>–0.044<br>0.965 | 0.710<br>0.637 | 0.290<br>0.363      | 4.913<br>6.256   | 0.016**<br>0.007**      |
| Overall (0–100)<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>$t_{58}$<br>$p$ -Value <sup>a</sup>     | 63.4 ± 14.7<br>67.4 ± 14.0<br>–4.0 (3.7)<br>–11.4, 3.4<br>–1.081<br>0.284 | 49.9 ± 21.9<br>44.9 ± 20.4<br>5.0 (6)<br>–7.1, 17.0<br>0.830<br>0.411   | 40.0 ± 25.9<br>40.2 ± 25.7<br>–0.1 (7.3)<br>–14.8, 14.5<br>–0.020<br>0.984 | 0.615<br>0.480 | 0.385<br>0.520      | 7.501<br>11.936  | 0.003**<br><0.001***    |

Abbreviations: ANOVA, analysis of variance; ANCOVA, analysis of covariance; CI, confidence interval; SD, standard deviation; SE, standard error.

<sup>a</sup>Unpaired *t* tests;  $t_{58}$ : *t* score at 58 degrees of freedom; <sup>b</sup>One-way repeated measure analysis of variance;  $p < 0.05$  considered as statistically significant; \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ ;  $p$ -value <sup>a</sup> reflects inter-group differences detected by unpaired *t*-tests, whereas  $p$ -value<sup>b</sup> represents intra-group changes detected by one-way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to  $R^2$  from multiple linear regressions.

$n = 60$ ; centesimal group: 30, 50-millesimal group: 30.

Adjusted for baseline differences in number of lesions by ANCOVA model; per protocol analysis.

**Table 6** Comparison of the DLQI scores at baseline and over 6 and 12 weeks

| DLQI scores                                                                                                                                           | Baseline:<br>Mean (SD)                                               | After 6 weeks:<br>Mean (SD)                                          | After 12 weeks:<br>Mean (SD)                                          | Wilks' lambda  | Partial eta squared | F <sub>2, 28</sub> | p-Value <sup>b</sup>   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|----------------|---------------------|--------------------|------------------------|
| Symptom and feeling<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup> | 5.3 ± 1.3<br>5.4 ± 1.0<br>-0.1 (0.3)<br>-0.7, 0.5<br>-0.437<br>0.664 | 3.3 ± 1.8<br>3.4 ± 1.8<br>-0.1 (0.5)<br>-1.0, 0.8<br>-0.218<br>0.828 | 2.5 ± 2.2<br>2.7 ± 2.5<br>-0.2 (0.6)<br>-1.4, 1.0<br>-0.386<br>0.701  | 0.417<br>0.427 | 0.583<br>0.573      | 19.557<br>18.801   | <0.001***<br><0.001*** |
| Daily activities<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>    | 4.9 ± 1.7<br>4.5 ± 1.8<br>0.4 (0.4)<br>-0.5, 1.3<br>0.818<br>0.417   | 2.7 ± 1.8<br>3.0 ± 2.2<br>-0.3 (0.5)<br>-1.4, 0.8<br>-0.572<br>0.570 | 1.9 ± 2.3<br>2.4 ± 2.5<br>-0.5 (0.6)<br>-1.7, 0.7<br>-0.806<br>0.423  | 0.420<br>0.623 | 0.580<br>0.377      | 19.366<br>8.486    | <0.001***<br>0.001**   |
| Leisure<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>             | 2.6 ± 1.2<br>2.4 ± 1.2<br>0.3 (0.3)<br>-0.3, 0.9<br>0.869<br>0.388   | 1.5 ± 1.3<br>1.4 ± 1.1<br>0.1 (0.3)<br>-0.5, 0.7<br>0.253<br>0.750   | 1.0 ± 1.4<br>1.3 ± 0.3<br>-0.3 (0.4)<br>-1.1, 0.5<br>-0.783<br>0.437  | 0.509<br>0.710 | 0.491<br>0.290      | 13.506<br>5.730    | <0.001***<br>0.008**   |
| Work and school<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>     | 2.6 ± 1.0<br>2.3 ± 1.3<br>0.3 (0.3)<br>-0.3, 0.9<br>0.992<br>0.325   | 1.4 ± 1.5<br>1.1 ± 1.5<br>0.3 (0.4)<br>-0.5, 1.1<br>0.776<br>0.441   | 1.1 ± 1.5<br>1.3 ± 1.5<br>-0.2 (0.4)<br>-1.0, 0.6<br>-0.519<br>0.605  | 0.494<br>0.600 | 0.506<br>0.400      | 14.318<br>9.333    | <0.001***<br>0.001**   |
| Personal relation<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>   | 2.6 ± 1.6<br>2.2 ± 1.7<br>0.3 (0.4)<br>-0.5, 1.2<br>0.780<br>0.438   | 1.5 ± 1.3<br>1.4 ± 1.3<br>0.1 (0.3)<br>-0.5, 0.8<br>0.404<br>0.688   | 1.0 ± 1.3<br>1.2 ± 1.6<br>-0.2 (0.4)<br>-0.9, 0.6<br>-0.434<br>0.666  | 0.584<br>0.772 | 0.416<br>0.228      | 9.956<br>4.131     | 0.001**<br>0.027*      |
| Treatment<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>           | 1.6 ± 1.5<br>1.6 ± 1.3<br>0.0 (0.4)<br>-0.7, 0.7<br>0.000<br>1.000   | 1.0 ± 1.1<br>1.0 ± 1.1<br>0.0 (0.3)<br>-0.6, 0.6<br>0.000<br>1.000   | 0.7 ± 1.1<br>1.0 ± 1.3<br>-0.3 (0.3)<br>-0.9, 0.3<br>-1.095<br>0.278  | 0.718<br>0.857 | 0.282<br>0.143      | 5.488<br>2.334     | 0.010*<br>0.116        |
| DLQI total<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>          | 19.5 ± 6.8<br>18.4 ± 6.8<br>1.1 (1.7)<br>-2.4, 4.6<br>0.649<br>0.519 | 11.4 ± 7.5<br>11.3 ± 8.2<br>0.1 (2.0)<br>-3.9, 4.2<br>0.066<br>0.948 | 8.1 ± 9.2<br>9.9 ± 10.3<br>-1.7 (2.5)<br>-6.8, 3.3<br>-0.687<br>0.495 | 0.478<br>0.596 | 0.522<br>0.404      | 15.305<br>9.501    | <0.001***<br>0.001**   |

Abbreviations: ANOVA, analysis of variance; ANCOVA, analysis of covariance; CI, confidence interval; DLQI, dermatological life quality index; SD, standard deviation; SE, standard error.

<sup>a</sup>Unpaired t-tests; t<sub>58</sub>: t score at 58 degrees of freedom; <sup>b</sup>One-way repeated measure analysis of variance;  $p < 0.05$  considered as statistically significant; \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ ; p-value a reflects inter-group differences detected by unpaired t-tests, whereas p-value<sup>b</sup> represents intra-group changes detected by one-way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to  $R^2$  from multiple linear regressions.

$n = 60$ ; centesimal group: 30, 50-millesimal group: 30.

Adjusted for baseline differences in number of lesions by ANCOVA model; intention to treat analysis.

total scores of DLQI questionnaire (►Table 6). In PP analysis, significant intra-group improvement (all  $p < 0.05$ , one-way repeated measure ANOVA) was found over time in all the subscale scores (except 50-millesimal scale of personal relation and treatment domain) and total scores of DLQI questionnaire (►Table 7).

Reductions of 50% or above in intensity of pruritus were found in facial (100%), hand (75%), breast (66.7%), abdomen (57.1%) and axillary (50%) lesions in the centesimal group and hands (80%), abdominal and facial (50% each) lesions in 50-millesimal group; thus, responses to treatment in lesions of hand were seemingly higher than other sites. Sixty-five per



**Table 7** Comparison of the DLQI scores at baseline and over 6 and 12 weeks

| DLQI scores                                                                                                                                           | Baseline:<br>Mean (SD)                                               | After 6 weeks:<br>Mean (SD)                                          | After 12 weeks:<br>Mean (SD)                                          | Wilks' lambda  | Partial eta squared | F <sub>2, 28</sub> | p-Value <sup>b</sup>   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|----------------|---------------------|--------------------|------------------------|
| Symptom and feeling<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup> | 5.3 ± 1.3<br>5.4 ± 1.0<br>-0.1 (0.3)<br>-0.7, 0.5<br>-0.437<br>0.664 | 3.0 ± 1.7<br>2.9 ± 1.6<br>0.1 (0.5)<br>-0.8, 1.1<br>-0.264<br>0.793  | 2.1 ± 2.1<br>2.0 ± 2.3<br>0.1 (0.6)<br>-1.1, 1.3<br>0.182<br>0.856    | 0.363<br>0.250 | 0.637<br>0.750      | 21.052<br>33.063   | <0.001***<br><0.001*** |
| Daily activities<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>    | 4.9 ± 1.7<br>4.5 ± 1.8<br>0.4 (0.4)<br>-0.5, 1.3<br>0.818<br>0.417   | 2.4 ± 1.7<br>2.5 ± 2.1<br>-0.1 (0.5)<br>-1.2, 0.9<br>-0.291<br>0.772 | 1.5 ± 2.1<br>1.8 ± 2.3<br>-0.3 (0.6)<br>-1.6, 0.9<br>-0.528<br>0.600  | 0.464<br>0.581 | 0.536<br>0.419      | 13.858<br>7.947    | <0.001***<br>0.003**   |
| Leisure<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>             | 2.6 ± 1.2<br>2.4 ± 1.2<br>0.3 (0.3)<br>-0.3, 0.9<br>0.869<br>0.388   | 1.3 ± 1.1<br>1.2 ± 1.1<br>0.1 (0.3)<br>-0.5, 0.7<br>0.321<br>0.750   | 0.7 ± 1.1<br>1.1 ± 1.6<br>-0.4 (0.4)<br>-1.1, 0.4<br>-0.905<br>0.370  | 0.479<br>0.696 | 0.521<br>0.304      | 13.066<br>4.794    | <0.001***<br>0.019*    |
| Work and school<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>     | 2.6 ± 1.0<br>2.3 ± 1.3<br>0.3 (0.3)<br>-0.3, 0.9<br>0.992<br>0.325   | 1.2 ± 1.5<br>0.9 ± 1.4<br>0.3 (0.4)<br>-0.5, 1.1<br>0.682<br>0.498   | 0.8 ± 1.3<br>1.1 ± 1.5<br>-0.3 (0.4)<br>-1.1, 0.5<br>-0.790<br>0.433  | 0.489<br>0.650 | 0.511<br>0.350      | 12.531<br>5.934    | <0.001***<br>0.009**   |
| Personal relation<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>   | 2.6 ± 1.6<br>2.2 ± 1.7<br>0.3 (0.4)<br>-0.5, 1.2<br>0.780<br>0.438   | 1.5 ± 1.2<br>1.3 ± 1.0<br>0.2 (0.3)<br>-0.5, 0.8<br>0.522<br>0.604   | 0.8 ± 1.2<br>1.0 ± 1.5<br>-0.2 (0.4)<br>-0.9, 0.6<br>-0.394<br>0.695  | 0.624<br>0.817 | 0.376<br>0.183      | 7.234<br>2.458     | 0.003**<br>0.109       |
| Treatment<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>           | 1.6 ± 1.5<br>1.6 ± 1.3<br>0.0 (0.4)<br>-0.7, 0.7<br>0.000<br>1.000   | 1.1 ± 1.1<br>0.8 ± 1.0<br>0.3 (0.3)<br>-0.4, 0.8<br>0.818<br>0.418   | 0.6 ± 1.0<br>0.8 ± 1.2<br>-0.2 (0.3)<br>-0.8, 0.5<br>-0.436<br>0.665  | 0.738<br>0.916 | 0.262<br>0.084      | 4.268<br>1.008     | 0.026*<br>0.381        |
| DLQI total<br>■ Centesimal group<br>■ 50-millesimal group<br>Mean group difference (SE)<br>95% CI<br>t <sub>58</sub><br>p-Value <sup>a</sup>          | 19.5 ± 6.8<br>18.4 ± 6.8<br>1.1 (1.7)<br>-2.4, 4.6<br>0.649<br>0.519 | 10.4 ± 7.2<br>9.6 ± 7.5<br>0.8 (2.1)<br>-3.4, 4.9<br>0.364<br>0.717  | 6.6 ± 8.5<br>7.8 ± 10.0<br>-1.2 (2.6)<br>-6.4, 4.1<br>-0.450<br>0.655 | 0.485<br>0.535 | 0.515<br>0.465      | 12.759<br>9.578    | <0.001***<br>0.001**   |

Abbreviations: ANOVA, analysis of variance; ANCOVA, analysis of covariance; CI, confidence interval; DLQI, Dermatological life quality index; SD, standard deviation; SE, standard error.

<sup>a</sup>Unpaired t-tests; t<sub>58</sub>: t score at 58 degrees of freedom; <sup>b</sup>One-way repeated measure analysis of variance; p < 0.05 considered as statistically significant; \*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001; p-value<sup>a</sup> reflects inter-group differences detected by unpaired t-tests, whereas p-value<sup>b</sup> represents intra-group changes detected by one-way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to R<sup>2</sup> from multiple linear regressions.

n = 60; centesimal group: 30, 50-millesimal group: 30.

Adjusted for baseline differences in number of lesions by ANCOVA model; per protocol analysis.

cent of patients whose duration of illness was 10 weeks or less than that, both the groups taken together, had reductions of 50% or above in intensity of pruritus. Similar development was noted in 50 and 33% of patients, when the duration of illness was 11 to 20 weeks and 21 to 30 weeks respectively. Hundred per cent reduction in pruritus was observed after 12 weeks of treatment in 27.6 and 22.2% of patients, when their duration of illness was 0 to 10 weeks and 11 to 20 weeks respectively. In

50-millesimal group, 56.7% patients were improved by the end of 6th week of treatment, while 66.7% patients were improved by the end of 12th week. Likewise, in centesimal group, 63.3 and 73.3% patients were improved by the end of 6th week and 12th week respectively.

Medicines prescribed: Sixteen medicines were prescribed in both the groups at baseline. Among those, *Natrum sulphuricum* (n = 17; 28.3%), *Natrum muriaticum* (n = 15; 25%), *Sulphur*

**Table 8** List of medicines prescribed in the two groups at baseline

| Sl. no. | Name of the medicines       | Total (n = 60); n (%) | Centesimal gr. (n = 30); n (%) | 50-millesimal gr. (n = 30); n (%) | Chi-squared test | p-Value <sup>a</sup> |
|---------|-----------------------------|-----------------------|--------------------------------|-----------------------------------|------------------|----------------------|
| 1.      | <i>Natrum sulphuricum</i>   | 17 (28.3)             | 6 (20)                         | 11 (36.7)                         | 1.313            | 0.252                |
| 2.      | <i>Natrum muriaticum</i>    | 15 (25)               | 9 (30)                         | 6 (20)                            | 0.356            | 0.551                |
| 3.      | <i>Sulphur</i>              | 5 (8.3)               | 1 (3.3)                        | 4 (13.3)                          | 0.873            | 0.350                |
| 4.      | <i>Bacillinum</i>           | 4 (6.7)               | 4 (13.3)                       | 0 (0.0)                           | –                | –                    |
| 5.      | <i>Pulsatilla nigricans</i> | 3 (5)                 | 2 (6.7)                        | 1 (3.3)                           | 0.000            | 1.000                |
| 6.      | <i>Mercurius solubilis</i>  | 2 (3.3)               | 1 (3.3)                        | 1 (3.3)                           | 0.517            | 0.472                |
| 7.      | <i>Rumex crispus</i>        | 2 (3.3)               | 1 (3.3)                        | 1 (3.3)                           | 0.517            | 0.472                |
| 8.      | <i>Sepia succus</i>         | 2 (3.3)               | 1 (3.3)                        | 1 (3.3)                           | 0.517            | 0.472                |
| 9.      | <i>Tellurium</i>            | 2 (3.3)               | 2 (6.7)                        | 0 (0.0)                           | –                | –                    |
| 10.     | <i>Thuja occidentalis</i>   | 2 (3.3)               | 0 (0.0)                        | 2 (6.7)                           | –                | –                    |
| 11.     | <i>Arsenicum album</i>      | 1 (1.7)               | 1 (3.3)                        | 0 (0.0)                           | –                | –                    |
| 12.     | <i>Calcarea carbonicum</i>  | 1 (1.7)               | 0 (0.0)                        | 1 (3.3)                           | –                | –                    |
| 13.     | <i>Capsicum annum</i>       | 1 (1.7)               | 1 (3.3)                        | 0 (0.0)                           | –                | –                    |
| 14.     | <i>Medorrhinum</i>          | 1 (1.7)               | 0 (0.0)                        | 1 (3.3)                           | –                | –                    |
| 15.     | <i>Silicea terra</i>        | 1 (1.7)               | 1 (3.3)                        | 0 (0.0)                           | –                | –                    |
| 16.     | <i>Tuberculinum bovinum</i> | 1 (1.7)               | 0 (0.0)                        | 1 (3.3)                           | –                | –                    |

<sup>a</sup>Pearson's chi-squared test,  $p < 0.05$  considered as statistically significant.

( $n = 5$ ; 8.3%) and *Bacillinum* ( $n = 4$ ; 6.7%) were the most frequently prescribed medicines in both the groups. The percentages of these medicines were differing in two groups (► **Table 8**) (Suppl. file 5: Indications of the prescribed medicines).

**Adverse events:** No harm, unintended effect, homeopathic aggravation, adverse event or any serious adverse events were reported from either group during the intervention period.

## Discussion

This open-label, two parallel arms, pilot RCT was performed on 60 patients suffering from TC and were treated with medicines in either 50-millesimal ( $n = 30$ ) or centesimal potency ( $n = 30$ ). Itching NRS was taken as the primary outcome, and Skindex-29 and DLQI were the secondary outcomes—all measured at baseline and at the intervals of 6th and 12th weeks. In both the groups, intra-group changes in the outcomes revealed significant improvement. There was no significant inter-group difference over 6th and 12th weeks; thus, indicating similar pattern of improvement in both the groups and both appeared equally effective. *Natrum sulphuricum*, *Natrum muriaticum*, *Sulphur* and *Bacillinum* were the most frequently prescribed medicines. Following the observations, an adequately powered definitive trial seemed to be feasible in future.

In this trial, the patients were assigned into the groups randomly from the earlier prepared random number chart; thus, the chances of selection bias were reduced. Enrolment in the study was prospective; that is, the protocol was declared prior to the enrolment of the first patient. The methodological strengths of the study also included the use of standardised and validated outcome measures. The sam-

ple size was too small to arrive at any definite conclusion regarding the therapeutic measure. Duration of follow-up might also be considered as inadequate in the context of usual recurrence of the symptoms.

Few case reports have been published till date claiming the usefulness of homeopathic medicines in TC.<sup>19–21</sup> But no studies have yet been performed to compare the effects of different scales of potencies in tinea infection. A clinical study evaluated the usefulness of *Bacillinum* in different potencies to verify Dr. Burnett's claim that tinea was curable by its pathological similimum in high potencies.<sup>11</sup> *Bacillinum* was found to be one of the most frequently indicated medicines in our study, but was prescribed on the basis of individualisation instead of empirically. In earlier homeopathic literature, we can find several cases of tinea treated and cured with homeopathic medicines.<sup>22,23</sup> None of the outcome of these cases was assessed by any validated scales, and all of them are retrospective case studies. In some cases of ringworm, which had been presented as pustular and encased with mattery scabs, Dr. J. C. Burnett had claimed that 'vaccinosis' may have barred the way to cure.<sup>23</sup> This 'vaccinosis' theory may also be studied using proper methodology. Few medicines are also suggested in older literature to use orally (like, *Sepia*, *Tellurium*, *Staphysagria*, *Rhus toxicodendron*, *Sulphur*, *Tabacum*, *Thuja*, *Bacillinum*) as well as locally (*Carbolic acid*, *Chloral*, *Tincture iodine*). These medicines are also subjected to be verified by performing good quality clinical studies.<sup>22–24</sup>

Future trials need to implement processes to account for the recruitment difficulties. A systematic recruitment strategy including adequate referrals from colleagues and from other outpatient departments and ensuring proper publicising about the trial may be helpful. Verbal reminder to

each referring practitioner, regular WhatsApp message reminders during clinical hours, posters in the waiting area and on the walls of outpatients were continued throughout the study period and those were useful to some extent. Patients were given reminders via phone calls just before the next scheduled follow-up visit to minimise attrition rates. From our study, a group of medicines emerged as the most frequently indicated ones. These medicines can be used in future trials. The English versions of the scales used to assess the outcomes were already validated; however, the validation studies of Bengali versions of these scales are yet to be published.

## Conclusion

Although certain recruitment difficulties were identified, an adequately powered trial exploring effectiveness of both scales in TC appeared to be feasible in future. Though the sample size was too small to arrive at any definite conclusion, both centesimal and 50-millesimal potencies of IH medicines appeared to be equally useful in reducing pruritus intensity and improving the quality of life in patients suffering from TC.

### Author Contribution

PH, SKM and SG were involved in concept, literature search, clinical study, data acquisition, data interpretation and preparation of the article; ASR and SD were involved in clinical study and data acquisition; SS, MK and S. Saha were involved in design, data interpretation, statistical analysis and preparation of the article. All the authors reviewed and approved the final article.

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### Conflict of Interest

We declare no conflict of interest. The trial was performed as the postgraduate thesis of the corresponding author.

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